# Drug-Target Interaction Mining

2026-04-17

## Introduction

Drug-target interaction data – which compounds bind which proteins at what affinity – underpins target discovery and drug repositioning. Public databases (ChEMBL, DrugBank, PubChem, BindingDB, PharmGKB) archive millions of bioassay measurements; programmatic access lets users assemble custom interaction tables.

## Prerequisites

Familiarity with compound identifiers (SMILES, InChI, ChEMBL ID) and protein identifiers (UniProt).

## Theory

Key bioassay quantities: - **IC50** – inhibitor concentration halving a biological response. - **Ki** – equilibrium inhibition constant. - **Kd** – dissociation constant. - **EC50** – effect concentration (agonist activity).

Lower = more potent. Values are typically reported in nM or uM; low-nM and sub-nM bindings are considered high-affinity drug-like hits.

## Assumptions

Assays are comparable across studies (unreliable in practice – format, conditions, species differ); target and compound annotations are correct.

## R Implementation

```r
library(httr); library(jsonlite); library(dplyr)

# Query ChEMBL's REST API for bioactivities of a target (ERBB2 / HER2 = CHEMBL1824)
# This illustrative snippet is internet-free pseudo-code:
chembl_url <- paste0("https://www.ebi.ac.uk/chembl/api/data/activity.json",
                     "?target_chembl_id=CHEMBL1824&limit=100")
# res <- fromJSON(content(GET(chembl_url), "text"))
# activities <- res$activities

# Instead, simulate a small bioactivity table for illustration
set.seed(2026)
df <- data.frame(
  compound = paste0("CMP", 1:200),
  target   = sample(c("HER2", "EGFR", "MET"), 200, replace = TRUE),
  type     = sample(c("IC50", "Ki", "Kd"), 200, replace = TRUE),
  value_nM = round(10^runif(200, 0, 4))
)

# Potent binders (< 100 nM) per target
df %>%
  filter(value_nM < 100) %>%
```

```r
  count(target, type, sort = TRUE)

# Rank compounds by median potency across multiple assays
df %>%
  group_by(compound, target) %>%
  summarise(median_pot = median(value_nM), n = n(), .groups = "drop") %>%
  arrange(median_pot) %>%
  head(5)
```

### Output & Results

A ranked table of potent drug-target interactions per target and per compound. In practice this replaces the simulated data with a ChEMBL pull of $>$ 10,000 measurements.

### Interpretation

"ChEMBL mining identified 26 HER2 inhibitors with IC50 $<$ 10 nM across independent assays; the most potent (CMP_X, median IC50 $=$ 2 nM) was selected for in-silico docking."

### Practical Tips

- Aggregate across assays using median or geometric mean to smooth inter-lab variance.
- Convert IC50 to pIC50 ($-\log_{10}$) for symmetric, Gaussian-like distributions.
- Filter by assay confidence score when using ChEMBL (confidence $\geq$ 7).
- Use `BindingDB` for affinity-focused data; `DrugBank` for approved-drug information.
- Cross-reference compounds via InChIKey to merge records across databases (SMILES is not unique).

### Reporting

A reproducible drug-target-interaction analysis names the database snapshot, the date of access, the activity types retained, and the confidence or quality filter applied to the raw assay records. State whether replicate measurements on the same compound-target pair were aggregated by median, mean of pIC50, or geometric mean, since these choices materially shift downstream rankings. When potency thresholds such as the conventional sub-100-nanomolar cutoff are used to define hits, justify the threshold in terms of the assay distribution rather than treating it as a universal constant, and report how compounds with mixed agonist or antagonist annotations were resolved before the ranking step.

### For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

### See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat